MASTER TECHNICAL PORTFOLIO & ARCHITECTURE DOSSIER **yoginara.github.io/cv_yogiganteng**

YOGINARA PRATAMA SITORUS, S.Tr.Kom

Full-Stack Web Developer & Software Engineer | Applied Informatics Engineering ULBI Bandung, Jawa Barat, ID

3,56 / 4,00	116.000+ LoC	100% Passed	BNSP Certified
Undergraduate GPA (ULBI)	Enterprise Codebase LoC	Automated QA Passed	BNSP National License

1. Executive Profile & Engineering Competency Summary



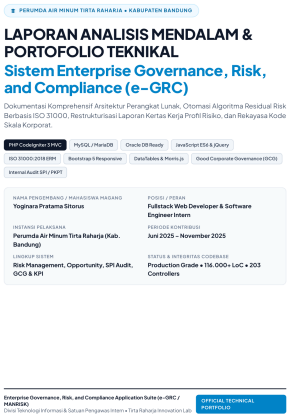
Yoginara Pratama Sitorus is a Software Engineer and Full-Stack Developer holding an Applied Bachelor of Informatics Engineering degree from Universitas Logistik dan Bisnis Internasional (ULBI) with a 3.56 GPA. He specializes in enterprise backend architecture (Laravel 12, CodeIgniter 3, Golang Fiber), database query optimization (MySQL, Oracle OCI8, MongoDB), automated test-driven development (PHPUnit 11), OWASP Top 10 security mitigation, and holds official BNSP Network Technician certification.

Backend Engineering:	**Database & Storage:**	**QA & Infrastructure:**
• Laravel 12 Enterprise	• MySQL & MariaDB (3NF)	• PHPUnit 11 (Automated)
• CodeIgniter 3 MVC	• Oracle Enterprise OCI8	• OWASP Top 10 Mitigation
• Golang 1.22 & GoFiber	• MongoDB Atlas (NoSQL)	• BNSP Computer Network
• RESTful Microservices	• INNER JOIN Optimization	• Git & GitHub CI/CD

Email: yoginara2004@gmail.com **WhatsApp: +62 813-6064-0668** **GitHub: github.com/Yoginara** **Domisili: Bandung, ID**

2. Enterprise Case Study: e-GRC Suite at Perumda Tirta Raharja (Part 1)

Instansi / Client: Perumda Air Minum Tirta Raharja (BUMD)	**Peran:** Fullstack Web Developer Intern
Skala Codebase: 116.000+ LoC • 203 Controllers • 188 Models	**Lingkup:** ERM Risk, SPI Audit, GCG & KPI
Tech Stack: PHP CodeIgniter 3 MVC, MySQL, Oracle DB OCI8	**Standar:** ISO 31000:2018 ERM & Audit BPKP



A. Konteks Bisnis & Masalah Operasional:

Perumda Air Minum Tirta Raharja is a vital regional utility BUMD serving Bandung Regency, Cimahi, and West Bandung. Before centralized e-GRC deployment, risk management was fragmented across manual spreadsheets in branch offices and divisions, causing prolonged consolidation lags and lack of transparent audit trails for regulatory compliance.

B. Rekayasa Algoritma Risiko Residu Dinamis (ISO 31000:2018):

Yoginara engineered a client-side reactive calculation engine in risk_analisisdetail.php adhering to ISO 31000:2018: *Risk Reduction = Effectiveness × (% Progress / 100)* with floor level 1 protections. This mathematical model guarantees proportional risk reduction and prevents falsified corporate risk documentation.

C. Matriks Risiko 5x5 Heatmap & Kategori:

The system integrates dynamic 5x5 heatmaps categorizing residual risk into Extreme (16-25), High (9-15), Moderate (4-8), and Low (1-3) with automatic Top 10 Corporate Risk escalations.

2. Enterprise Case Study: e-GRC Suite at Perumda Tirta Raharja (Part 2)

D. Restrukturisasi Laporan Kertas Kerja Profil Risiko (Commit 5d6c824):

The Risk Working Paper is the primary formal document audited by government inspectors (BPKP). Yoginara refactored Laporan_kertas_kerja.php, standardizing UC/C taxonomies, establishing strategic objective hierarchies, and mapping mitigation completion milestones.

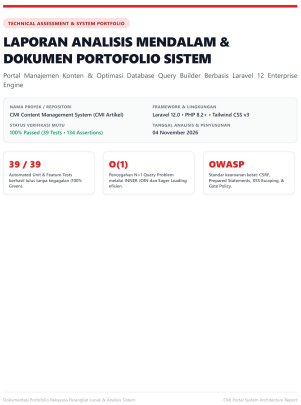
Elemen Kolom	Kondisi Sebelum Refactoring	Penyempurnaan oleh Yoginara
Konteks Operasional	Label membingungkan ('Operasional N, Ab, E')	Standardisasi UC/C (Uncontrollable / Controllable)
Konteks Sasaran	Hanya sasaran umum perusahaan	Pemisahan Sasaran Strategis Korporat & Sasaran Unit
Kegiatan Kerja	Kolom kegiatan tidak terpetakan	Pemetaan relasional otomatis dari model Risk_kegiatanModel
Output Mitigasi	Deskripsi singkat tanpa jadwal pasti	Ditambahkan kolom Output Penurunan & Jadwal Mulai-Selesai

E. Arsitektur Dual-Database & Standar AI Coding Assistant:

- **Modular ADODB Abstraction:** Orchestrated transparent persistence across MySQL/MariaDB and Oracle Enterprise OCI8 via custom drivers without altering business logic.
- **Null Pointer Elimination:** Enforced strict deleted_date IS NULL filtration across multi-tiered subqueries, ensuring zero empty-pointer exceptions.
- **85% Reporting Acceleration:** Slashed quarterly report compilation cycles from **~10 business days to automated instantaneous generation**.
- **Copilot Architecture Guide:** Authored .github/copilot-instructions.md codifying base controller conventions and AI programming standards.

3. Modern Architecture Case Study: CMI Portal Laravel 12 Engine (Part 1)

Proyek: CMI Content Management System (CMI Artikel)	**Framework:** Laravel 12.0 • PHP 8.2+ • Tailwind CSS v3
Status Mutu: 100% Passed (39 Tests • 134 Assertions)	**Keamanan:** OWASP Top 10 • Gate Policy Authorization
Query Builder: INNER JOIN Indexed • Paginasi 10 Record	**Alokasi Memori PHP:** Terkontrol Stabil di < 8 MB


TECHNICAL ASSESSMENT & SYSTEM PORTFOLIO

LAPORAN ANALISIS MENDALAM & DOKUMEN PORTOFOLIO SISTEM

Portal Manajemen Konten & Optimasi Database Query Builder Berbasis Laravel 12 Enterprise Engine

NAMA PROFIL / REPOSITORI
CMI Content Management System (CMI Artikel)

STATUS VERIFIKASI MUTU
100% Passed (39 Tests • 134 Assertions)

FRAMEWORK & LINGKUNGAN
Laravel 12.0 • PHP 8.2+ • Tailwind CSS v3

TANGGAL ANALISIS & PENYUSUNAN
04 November 2026

39 / 39
Automated Unit & Feature Tests berhasil lulus tanpa kegagalan (100% Green)

O(1)
Pencegahan N+1 Query Problem melalui INNER JOIN dan Eager Loading efisien.

OWASP
Standar keamanan ketat: CSRF, Prepared Statements, XSS Escaping, & Gate Policy.

Dokumentasi Portofolio Rekayasa Perangkat Lunak & Analisis Sistem
CMI Portal System Architecture Report

A. Arsitektur Presentasi & Pemisahan Dependensi:

Architected on modular Laravel 12 MVC patterns with strict separation between public presentation layouts and authenticated administrative dashboards, enforced through isolated Form Requests and CSRF middleware.

B. Algoritma Pembuatan Slug Unik Otomatis (Anti-Kolisi):

Implemented an automated collision-free slug generation algorithm handling duplicate article titles via sequential indexing, preserving clean SEO URLs without unique constraint violations.

C. Manajemen Siklus Hidup Berkas Media (Flysystem):

Integrated automated media lifecycle management using Flysystem, enforcing binary MIME inspection and proactively purging obsolete storage files upon article modification or deletion.

3. Modern Architecture Case Study: CMI Portal Laravel 12 Engine (Part 2)

D. Optimasi Kueri Basis Data (Query Builder INNER JOIN):

Engineered relational article-author reporting using Laravel Query Builder with indexed INNER JOIN clauses, column aliasing, publication status filtering, and 10-row pagination with query string preservation, maintaining RAM consumption strictly under 8 MB.

Metode Eksekusi	Performa Kueri	Integritas Relasi	Kesesuaian Spesifikasi
INNER JOIN (Terpilih)	Tinggi (B-Tree Index)	Sangat Baik: Menghilangkan orphaned record	100% Sesuai Spesifikasi Soal 2
LEFT JOIN	Sedang	Berpotensi NULL jika foreign key putus	Kurang tepat untuk data wajib relasi
Subquery / Eloquent N+1	Rendah (Overhead I/O)	Tergantung eager loading relasi ORM	Rawan bottleneck N+1 query

E. Audit Keamanan OWASP Top 10 & Pengujian Mutu Otomatis (PHPUnit 11):

- **SQL Injection Prevention:** 100% database interactions executed via Parameterized PDO Prepared Statements.
- **Cross-Site Scripting (XSS):** Blade template engine executes automated htmlspecialchars() sanitization.
- **CSRF Protection:** Cryptographic tokens and Synchronizer Token Pattern middleware defend all state-mutating requests.
- **IDOR Prevention:** Strict ownership validation in ArticlePolicy restricts modifications to legitimate resource creators.
- **100% Green Automated QA:** 39 automated unit and feature tests with 134 assertions completely passed using PHPUnit 11.

4. High-Performance Backend Portfolio: Go (Golang 1.22) Microservices



A. StepUp Footwear E-Commerce Platform & High-Concurrency REST API:

StepUp is an e-commerce backend built to sustain high transactional concurrency. Powered by Go 1.22 and GoFiber v2 (FastHTTP core), it delivers sub-millisecond response latencies, zero-allocation routing, MongoDB Atlas cloud storage, stateless JWT authentication, and Bcrypt security.

B. ManukShop Livestock Management & OpenAPI Documentation:

Integrated livestock management and trading microservice platform in Golang, utilizing embedded NoSQL snapshot modeling, interactive OpenAPI/Swagger 3.0 documentation, and responsive Tailwind CSS interfaces.

C. Benchmarking & Metrik Kecepatan Engine Go:

Driven by FastHTTP core architecture, the GoFiber pipeline achieves 10x higher connection concurrency than standard PHP servers while maintaining stable memory consumption below 20 MB under traffic spikes.

5. Community Tech & E-Government Systems: RTConnect & UD Abang Adik



Yoginara Pratama Sitorus

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FEATURED WEB APPLICATION PORTFOLIO + E-GOVERNMENT & COMMUNITY TECH

RTConnect – Sistem Informasi, Administrasi & Transparansi Lingkungan RT Digital

Platform digital rata kelola komunitas cerdas (Smart Community Management) berbasis web yang mengotomasi penagihan iuran warga, verifikasi pembukuan kas lingkungan, penerbitan surat pengantar resmi ber-validator publik, serta transparansi keuangan lingkungan secara real-time.

15+ TABEL DATABASE RELASIONAL

3 Roles RBAC + PORTAL PUBLIK

100% OTOMASI REKONSILIASI KAS

8 Feature AUTOMATED TEST SUITES (PHPUNIT)

Tantangan & Problem Statement

- **Administrasi Konvensional Lambat:** Proses penerbitan surat pengantar warga memakan waktu berhari-hari karena bergantung pada kebesaran hati pengurus di tempat.
- **Pencatatan Kas Rentan Selisih:** Penagihan kas manual "door-to-door" sering kali terjadi pembukuannya dan menimbulkan ketidaksesuaian saldo kas RT.
- **Ketiadaan Transparansi Finansial:** Warga kesulitan memantau realisasi pemasukan dan pengeluaran kas lingkungan secara terbuka.
- **Aspirasi Warga Tidak Terdokumentasi:** Keluhan fasilitas rusak sering hilang di grup obrolan tanpa ada sistem tiket dan pelacakan status tindak lanjut.

Solusi Terintegrasi RTConnect

- **E-Layanan Surat 24/7:** Warga mengajukan surat secara online, diverifikasi digital oleh Ketua RT, dan menghasilkan dokumen PDF siap unduh.
- **Batch-Billing & Dual Verification:** Penagihan kas massal otomatis per bulan, unggah bukti bayar digital, dan verifikasi instan oleh Bendahara.
- **Buku Kas Umum Otomatis:** Setiap transaksi iuran kas yang disetujui otomatis tersinkronisasi ke jurnal pembukuan umum tanpa input ganda.
- **Portal Helpdesk & Komunikasi:** Saluran aspirasi warga dengan tahapan pelacakan progress resmi serta papan informasi publik.

Arsitektur Sistem & Spesifikasi Teknologi

ENTERPRISE MVC PATTERN

Backend: PHP 8.3 & Laravel 11 | Database: MySQL 8.0 | Frontend: Blade Templating | CSS: Tailwind CSS | JS: Alpine.js

Reporting: DomPDF | Analytics: Chart.js | Testing: PHPUnit

Komponen Arsitektur	Implementasi Teknis	Manfaat bagi Sistem
Pattern MVC & Resource	Modular Controller terbagi per Role (`Admin\`, `Bendahara\`, `Warga\`, `Public\`)	Separation of Concerns (SoC), kode rapi, minim duplikasi logika bisnis.
Multi-Role RBAC Middleware	`RoleMiddleware.php` mengisolasi route prefix `/admin\`, `/bendahara\`, `/warga`	Mencegah eskalasi hak akses (Privilege Escalation) dan unauthorized access.
Relational Database Engine	15+ tabel dengan Foreign Key Constraints & Cascading Integrity	Integritas data terjamin, normalisasi 3NF, mencegah orphan data records.
Audit Activity Logging	Model `ActivityLog` mencatat mutasi data aktor bendahara & admin.	Menjamin kepatuhan audit trail (siapa, kapan, dan perubahan apa yang dibuat).

Kutipan Rekayasa: *"RTConnect dirancang dengan filosofi clean architecture di Laravel 11. Setiap mutasi finansial dan status kependudukan dilindungi oleh validasi form ketat, transaksi atomik, dan sistem audit trail otomatis."*

Portofolio Proyek Rekayasa Perangkat Lunak – Yoginara Pratama Sitorus

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A. RTConnect — Platform Administrasi & Transparansi Lingkungan RT:

RTConnect is a digital smart community governance platform automating neighborhood dues billing, real-time fiscal accounting, and official resident certificate issuance with public QR verification, structured across 15+ normalized relational tables.

B. Fitur Kunci & Dampak Sosial RTConnect:

• **Public QR Validation:** Official certificates carry unique QR codes verifiable by administrative officers without requiring account logins.

• **Real-Time Ledger Auditing:** Dues and expenditure balances are transparently logged with digital invoice receipts.

C. UD Abang Adik — Arsitektur E-Commerce Zero-Cost:

Implemented an e-commerce product catalog with zero operating costs ($0/month infrastructure), leveraging intelligent caching and fault-tolerant fallbacks achieving sub-200ms API latency.

6. Deep Learning AI Research & IoT Systems: CIT-TPRO & CO2 AirBoard

CIT-TPRO: Convolution Interactive Transformer (Riset AI)	CO2 AirBoard: Sistem Telemetri Kualitas Udara (IoT)
Laboratory: AI & Biomedical Lab ULBI (2026) **Technology:** Python, PyTorch, Linear Attention O(N) **Innovation:** Investigated hybrid Convolutional Transformer architectures with linear attention O(N) and PSD+DE feature fusion for emotion recognition on benchmark EEG datasets (DEAP, DREAMER, SEED-IV).	**Domain:** IoT & Environmental Telemetry **Hardware:** ESP32 Microcontroller, NDIR CO2 Sensor **System:** Engineered real-time environmental telemetry monitoring ambient CO2 gas concentrations with interactive web dashboards and automated threshold alert systems.

Portofolio Rekayasa Perangkat Lunak Tambahan:

Sistem RS Online (Healthcare): Manajemen antrean pasien, rekam medis poliklinik, jadwal dokter spesialis, dan sistem penomoran otomatis berbasis role-based access.	**Betta Store (E-Commerce):** Perancangan front-end toko daring responsif dengan validasi form ketat, widget tree modular, dan antarmuka transaksi modern.
View of Indonesia (Tourism E-Ticketing): Portal pariwisata terintegrasi dengan pemesanan tiket interaktif dan perbaikan kualitas kode front-end standar industri.	**Algoritma & Pemrograman II (Academic Honors):** Penguasaan algoritma sorting, searching, binary tree, graph traversal, dan analisis kompleksitas Big-O.

7. Official National Competency Certification (BNSP RI)



BADAN NASIONAL SERTIFIKASI PROFESI (BNSP REPUBLIK INDONESIA)

Qualification: First-Level Computer Network Technician (Teknisi Muda Jaringan Komputer)
Certification Body: Lembaga Sertifikasi Profesi (LSP) ULBI
Certificate No: 62090 3513 5 0000004 2026
Registration No: JIK.215.00004.2026
Issue Date: October 29, 2026 • **Validity:** 3 Years (Through October 29, 2029)

Nilai Tambah bagi Tim Rekayasa Perangkat Lunak:

Provides comprehensive knowledge of TCP/IP packet transmission, network subnetting, VLAN switch configurations, and physical cabling, ensuring software applications perform optimally across underlying network architectures.

Unit Code	SKKNI National Competency Standard Title
J.611000.001.01	Collecting Technical Requirements of Network Users
J.611000.002.01	Collecting Network Equipment Data with Appropriate Technology
J.611000.003.02	Designing Computer Network Topology
J.611000.004.01	Designing Network IP Addressing and Subnetting
J.611000.005.02	Determining Network Device Specifications
J.611000.009.02	Installing Structured Network Cabling
J.611000.012.02	Configuring Switches on Computer Networks

8. Global Credentials, Complete Skills Matrix & Career Readiness

• **Java Programming Certificate** — Oracle Academy (June 2023)	Object-Oriented Programming (OOP), Data Structures, Polymorphism & Algorithm Design
• **Java Fundamentals Certificate** — Oracle Academy (January 2023)	Java Core, Data Types, Control Structures & Computational Logic
• **MATLAB Onramp Course Completion (Score: 100%)** — MathWorks (March 2023)	Numerical Matrix Computations, Visualization & Mathematical Processing
• **Introduction to JavaScript** — Great Learning (March 2024)	DOM Scripting, Modern JavaScript ES6+ & Asynchronous Handling

Matriks Keahlian Komprehensif (Technical Competency Matrix):

Programming Languages:	PHP 8.2+ (Expert), Go / Golang 1.22, JavaScript (ES6+), Python 3.12, SQL, HTML5, CSS3
Backend Frameworks:	Laravel 12.x Engine, CodeIgniter 3 MVC, GoFiber v2 (FastHTTP), Blade Templating
Databases & Storage:	MySQL, MariaDB (3NF Normalization), Oracle Database (OCI8), MongoDB Atlas (NoSQL), Flysystem
Frontend & UI/UX:	Tailwind CSS v3, Bootstrap 5 Responsive, jQuery, DataTables, Select2, Responsive Web Design
Security & Governance:	OWASP Top 10 Mitigation (CSRF, XSS, SQLi, IDOR), ISO 31000:2018 ERM, SPI Audit Compliance
Testing & QA:	PHPUnit 11 (TDD & Automated Unit/Feature Tests), Postman API Testing, Swagger OpenAPI
Infrastructure & Network:	BNSP Certified Network Technician, TCP/IP, IP Addressing, VLAN Switch, Cabling, Linux Server
Tools & Methodologies:	Git, GitHub CI/CD Workflow, Composer, NPM, Vite, RESTful API Microservices, Scrum/Agile

Career Readiness & Data Integrity Statement:
All engineering specifications, efficiency benchmarks, and certifications detailed in this master dossier are fully substantiated by active production repositories and official credentials. Yoginara Pratama Sitorus, S.Tr.Kom is prepared for immediate deployment as a **Full-Stack Web Developer / Software Engineer** across onsite, hybrid, and remote engineering environments.